Endogenizing Nature in Production Theory: A Framework of Economic Commensalism

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Abstract

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Keywords: economic commensalism; environmental economics; rational incentives; commons; carbon markets.

1. Introduction

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The relationship between economic activity and the natural environment has long been a subject of scholarly and social inquiry, especially regarding the tension between the productive use of natural resources and ecological sustainability. Economic theory does not ignore nature. Natural resources have been examined as stocks, constraints, sources of land rent, externalities, and environmental costs. The problem addressed in this article is more specific: in conventional production theory, nature rarely appears as an endogenous, dynamic, and strategically managed factor capable of affecting future productivity, private incentives, and intertemporal welfare. Although land, capital, and labor are commonly

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treated as fundamental inputs of production, the ecological systems that sustain them are often modeled as external constraints, fixed stocks, or sources of externalities, rather than as dynamic components of the production process itself (JEHLE; RENY, 2011; STIGLITZ, 1989; DASGUPTA, 2001). This separation has important implications for environmental policy.

Environmental degradation is frequently treated as an ex post consequence of production decisions, to be corrected through instruments such as taxation (PIGOU, 2010), regulation, property-rights allocation (HARDIN, 1968), carbon pricing, or collective governance arrangements (OSTROM, 2002). Although these mechanisms have shaped much of modern environmental policy, they often affect agents through compliance costs, legal constraints, or compensatory payments, rather than by making ecological preservation a structurally endogenous component of production and profitability.

In practical terms, this separation is reflected in legal and institutional arrangements that impose preservation duties on private agents without fully integrating ecological preservation into their economic rationale. For example, an agricultural producer may be required to preserve part of a biome even when doing so reduces the amount of land available for production and, consequently, constrains productive capacity, as occurs in Brazil, where legislation restricts land use to varying degrees according to the biome (BRASIL, 2012). This requirement creates an opportunity cost: land that could otherwise be used productively must be preserved in the interest of protecting a collective good. The agent therefore faces a trade-off between production and preservation, often framed as a conflict between economic growth and environmental conservation, or as a public benefit obtained at the expense of private costs.

–BLOCO 1–

This limitation becomes clearer when one considers the main instruments currently used to reconcile production and environmental preservation. Positive incentives, such as carbon markets, payments for environmental services, environmental certifications, fiscal incentives, and production rights conditioned on conservation, are designed to reward agents for reducing environmental harm or maintaining ecological assets. Negative incentives, such as inspections, fines, embargoes, prohibitions, and legal restrictions, seek to deter environmentally harmful behavior by increasing its expected cost. Although these mechanisms may reduce degradation under specific conditions, they often operate as external corrections to production decisions rather than as internal components of the production process itself. A fine may increase the cost of illegal degradation; a subsidy may compensate the agent for preserving land; a carbon credit may create a tradable asset associated with reduced emissions; and a certification may increase market access

or generate a price premium. Yet, in each case, ecological preservation may remain an additional condition attached to production, rather than a productive factor whose maintenance directly affects expected profitability, productivity, and intertemporal strategy (**????**).

A second limitation concerns the spatial and institutional design of environmental compensation. Many environmental instruments are centralized, standardized, or globally tradable, while ecological degradation is territorially specific. A producer who degrades a local biome may compensate emissions or environmental impacts through mechanisms located elsewhere. This may satisfy an accounting or regulatory requirement, but it does not necessarily restore the ecological functions of the affected territory. The problem is therefore not only whether compensation occurs, but where it occurs and whether it preserves the ecological conditions that sustain local production, resilience, biodiversity, hydrological cycles, and long-term productivity. When the benefits of preservation are detached from the ecological unit affected by production, and when the costs of degradation can be offset through distant or standardized mechanisms, environmental compensation may become a price paid for the right to continue degrading locally. In such cases, environmental policy may reduce aggregate damage in formal terms while failing to align private incentives with the ecological systems upon which future production depends (**????**).

A third limitation is the lack of coordination between positive and negative incentives over time. In many institutional settings, environmental policy combines rewards and punishments without integrating them into a coherent intertemporal incentive structure. The agent may face legal restrictions, occasional enforcement, market opportunities, subsidies, reputational gains, certification requirements, and fiscal incentives, but these instruments often operate through different agencies, time horizons, eligibility criteria, and territorial scales. As a consequence, the expected payoff associated with preservation may remain uncertain, fragmented, or inferior to the immediate gains from expanding production or reducing compliance costs. This helps explain why preservation and productive use are frequently perceived as incompatible: in the short run, preservation may reduce available land, increase compliance costs, or limit operational flexibility, while its benefits are delayed, uncertain, spatially detached, or captured by society at large. However, this incompatibility is not a necessary feature of the relationship between production and preservation. It is, at least in part, a consequence of incentive calibration. If ecological preservation were incorporated as an endogenous component of production, and if its benefits were linked to local productivity, access to credit, market premiums, fiscal advantages, production rights, or long-term profitability, the agent's calculus could change. Preservation would no longer be merely a restriction on productive activity; it

could become a condition for maintaining or increasing expected returns across time.

This dichotomy can be understood through three recurring paradigms of human–environment interaction: predation, mutualism, and extractivism. Predation denotes the unsustainable and often illegal depletion of natural resources; mutualism refers here to a preservation-oriented ideal in which human use is strongly limited in favor of ecological integrity; and extractivism occupies an intermediate position, allowing resource use within supposedly sustainable or regenerative limits (GUDYNAS et al., 2009; ACOSTA, 2013). From an efficiency perspective, however, none of these paradigms fully resolves the production–preservation trade-off. Predation disregards ecological constraints and generates negative externalities, while strict mutualism may excessively restrict productive potential (DASGUPTA, 2001; OSTROM, 1990). Extractivism, in turn, may delay ecological collapse but remains vulnerable to regulatory failure: if too restrictive, it underuses productive resources; if too permissive, it reproduces the overuse problem it was meant to avoid (CHICHILNISKY, 2017). Therefore, natural resource allocation tends to deviate from Pareto efficiency whenever private incentives are not aligned with social and ecological costs, which motivates the search for institutional arrangements capable of reconciling productive efficiency with environmental sustainability (HANLEY et al., 2007).

Against this background, the central **problem** addressed in this article is whether environmental preservation can be theoretically reframed not merely as a constraint on production, but as an endogenous component of economic decision-making. More specifically, the article asks whether private productivity and profitability can be reconciled with socially desirable environmental preservation through incentive-compatible arrangements capable of overcoming the limitations of the prevailing extractivist logic.

The **hypothesis** advanced here is that, when ecological variables are incorporated into production theory and when incentives are designed to reflect spatial, intertemporal, and behavioral constraints, environmental preservation may become compatible with individual rationality rather than dependent solely on coercion or moral commitment. In this sense, the article rests on four **premises**: first, the environment can be represented as an endogenous cofactor of production; second, many existing environmental instruments fail to alter the structural private payoff that drives degradation; third, preservation becomes stable only when it is compatible with individual rationality and strategic equilibrium; and fourth, such compatibility requires governance mechanisms that are locally anchored in ecological units, while remaining subject to broader systems of verification and institutional credibility.

The **objective** of the article is therefore to develop the conceptual foundations of a framework for endogenizing nature in production theory and for reinterpreting preserva-

tion as a potentially rational, incentive-compatible, and productivity-enhancing economic strategy.

2. The Limits of Current Environmental Governance

—> Mecanismos atuais de governança ambiental (regulação, precificação, mercados de carbono, etc.) são insuficientes para lidar com a complexidade e a dinâmica dos sistemas socioecológicos.

—> Dados e evidências mostram que, apesar dos esforços, as tendências de degradação ambiental e perda de biodiversidade continuam, indicando falhas estruturais nos modelos de governança atuais.

—> Citar esforços internacionais (Acordo de Paris, Objetivos de Desenvolvimento Sustentável) e nacionais (políticas ambientais, leis de proteção) que não têm conseguido reverter as tendências negativas.

—> Usar dados de desmatamento, perda de biodiversidade, emissões de gases de efeito estufa, Brasil e mundo, para ilustrar a insuficiência dos mecanismos atuais.

–> Concluir que é necessário repensar os fundamentos teóricos e práticos da governança ambiental, buscando abordagens que integrem melhor a natureza como parte dos sistemas econômicos e sociais, e que sejam mais adaptativas, inclusivas e eficazes.

3. Literature Review: From Externalities to Incentives

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4. Model

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5. Results

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6. Discussion

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7. Final Remarks

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